

Unit 2, Lesson 3: Rewriting Expressions Using the Distributive Property

Benchmark

MA.6.NSO.3.2 Rewrite the sum of two composite whole numbers having a common factor, as a common factor multiplied by the sum of two whole numbers.

Learning Target

- ☐ I can use the distributive property to generate equivalent numeric expressions.

2.3.1 Warm-Up: How Many Ways?

1. Without using the number 1 as a factor, determine different ways the number 120 can be written as a product of three factors. Find at least three different expressions.
2. Work with your partner to share the different ways 120 can be expressed as a product of three factors. Add any new expressions shared to your list above and discuss similarities and differences between the expressions you and your partner determined.

2.3.2 Exploration: The Distributive Property

A local grocery store manager is trying to determine how many soda cans she will need to make a new display. The specifications require a total of 7 rows with 23 cans in each row.

To determine the number of cans needed, she multiplies the number of rows by the number of cans in each row.

1. What is the value of 7×23 ? Show your work.
2. Compare your work with two other students.

Name	Did you get the same answer?	Did you use a similar method? If not, summarize their method below.

- 3.
- a. Find the products.

7×20

7×3

- b. What is the sum of the two products?

4. How does the sum of those two products compare to the product of 7×23 ?

The sum of $(7 \times 20) + (7 \times 3)$ the product of 7×23 .

☐ is less than

☐ is equal to

☐ is greater than

Area models can be helpful when finding products of numbers. The figure to the right can be used to model the product of 7×23 .	23 7
The number 23 can also be decomposed into $(20 + 3)$ to find the same product. In this area model, both parts of the area model can be added to find the total area.	20 + 3 7

5. How can decomposing the number 23 into $20 + 3$ make multiplying easier?

2.3.3 Guided Instruction: The Distributive Property



The **distributive property of multiplication over addition** is a property that applies to all rational numbers. The distributive property “distributes” a factor to all terms within a set of parentheses.

$$a \times (b + c) = (a \times b) + (a \times c)$$

The product 7×23 can be rewritten using the distributive property by decomposing 23 into two parts and multiplying 7 by both parts.

$$\begin{aligned} 7 \times 23 &= 7(20 + 3) \\ 7(20 + 3) &= (7 \cdot 20) + (7 \cdot 3) \end{aligned}$$

1. What are two other ways that 7×23 could be rewritten using the distributive property?

$$7 \times 23 = \underline{\hspace{1cm}} (\underline{\hspace{1cm}} + \underline{\hspace{1cm}})$$

$$7(23) = \underline{\hspace{1cm}} (\underline{\hspace{1cm}} + \underline{\hspace{1cm}})$$

2. Which of the expressions is the easiest to evaluate using mental math?

The distributive property can also be used to rewrite the sum of two numbers that share a common factor.

3. Please rewrite the sum $27 + 18$ as $3(9 + 6)$. Are these two expressions equivalent?

a. $27 + 18$ and $3(9 + 6)$

☐ are

☐ are not

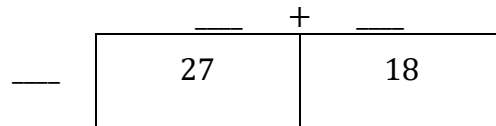
equivalent

expressions because when evaluated they have

☐ the same value.

☐ different

- b. What common factor of 27 and 18 did Elease use to rewrite the expression?
- c. Rewrite the expression $27 + 18$ using a different common factor of 27 and 18. Use the area model to support your thinking.



$$27 + 18 = \underline{\hspace{1cm}}(\underline{\hspace{1cm}} + \underline{\hspace{1cm}})$$

2.3.4 Guided Practice: Rewriting Expressions

1. Complete the table by rewriting each multiplication expression using the distributive property, modeling with area, and finding the product.

Multiplication Expression	Distributive Property	Area Model	Product
98×5	$5 (90 + 8)$	5 90 + 8 45040	
6×22			

2. Complete the table by rewriting each addition expression using the distributive property, modeling with area, and finding the sum.

Addition Expression	Distributive Property	Area Model	Sum
$600 + 140$		20 30 + 7 600140	
$56 + 32$			

2.3.5 Your Turn!

1. For each expression, generate an equivalent numeric expression using the distributive property and then find the value.

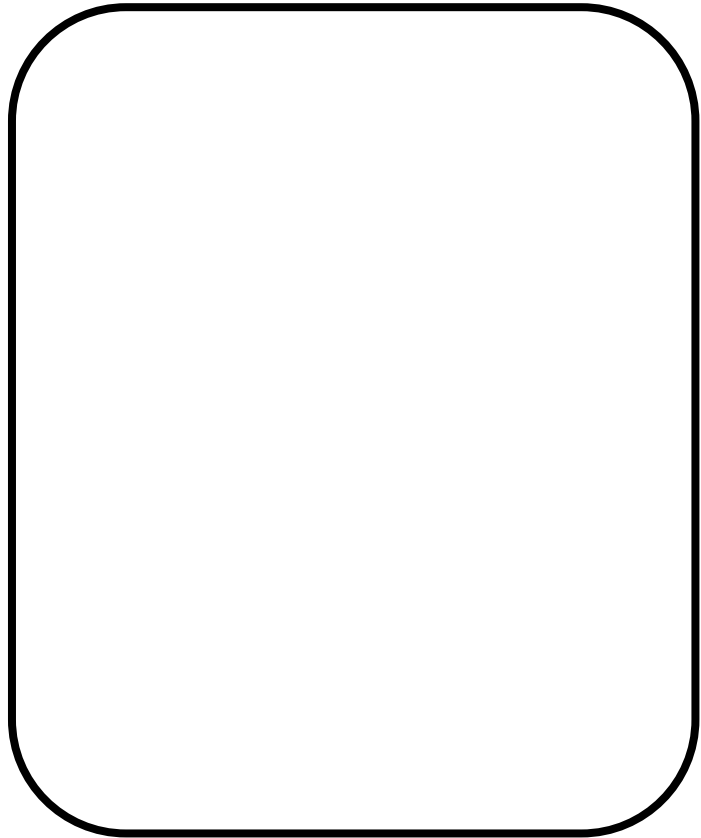
Expression	Equivalent Expression Using Distributive Property	Value
9×16		
$25 \cdot 8$		
39×4		
$48 + 72$		
$21 + 98$		

2. Bailey rewrote the expression $36 + 81$ using the distributive property, but he made an error.
- a. Evaluate his work and determine his mistake.

Given Expression	Bailey's Expression	Corrected Expression
$36 + 81$	$9(4 + 81)$	

- b. Help Bailey understand and learn from his mistakes. Write a brief note to Bailey explaining what his mistake was and how he can learn from it.

Mmeans
I
Sstart
To
Acquire
Knowledge
Experience
Skills



2.3.6 Wrap-Up: Growth Mindset

1. Complete each statement below based on your learning today.

Today I am still unsure about _____

_____.

To fill this gap, I plan to _____

_____.